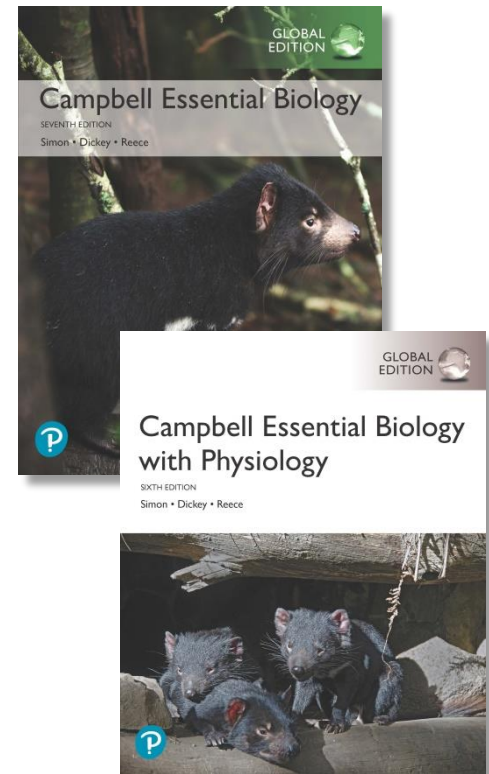


Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 22 Nutrition and Digestion



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Eat Your Carrots! Too Little **Vitamin A** can Jeopardize Your Vision, but Eating too Much can Turn Your Skin Orange



Biology and Society: The “Secret” to Shedding Pounds (1 of 2)

- Each year, about one in seven Americans starts a diet. But only about 5% of dieters are able to reach their goal weight and maintain it for the long term.
 - More than a third of American adults are obese (very overweight).
 - Obesity contributes to heart disease, diabetes, cancer, and other problems.

Biology and Society: The “Secret” to Shedding Pounds (2 of 2)

- There really is no trick to managing your weight.
 - Add up the calories from the food you eat. Subtract the calories that your body burns.
 - If you take in more than you burn, you will gain weight. If you burn more than you take in, you will lose weight.
 - Weight control can be summed up in a five-word “secret”: Eat less and exercise more!
 - But caloric balance alone does not ensure good nutrition. Food must also provide the raw materials for building healthy cells and tissues.

The Key to Maintaining a Healthy Weight



An Overview of Animal Nutrition

- As animals, we must eat other organisms to acquire nutrients.
 - Food provides the raw materials we need to build tissue and fuel cellular work.
 - However, food primarily consists of large, complex molecules that are not in a form an animal's cells can use.
 - Thus, the body must break down these nutrients—digest them—to make them useful.

Animal Diets (1 of 2)

- All animals eat other organisms, dead or alive, whole or by the piece. Beyond that generalization, however, animal diets vary extensively.
 - **Herbivores** feed mainly on plants and/or algae.
 - **Carnivores** mainly eat other animals.
 - **Omnivores** eat animals as well as plants and/or algae.

Animal Diets (2 of 2)

ANIMAL DIETS		
Herbivore (mainly eats plants and/or algae)	Carnivore (mainly eats animals)	Omnivore (regularly eats animals as well as plants and/or algae)
		

The Four Stages of Food Processing

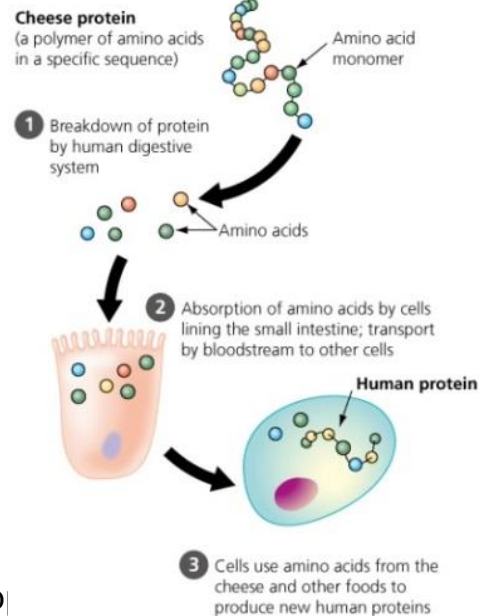
There are four stages of food processing.

1. **Ingestion** is another word for eating.
2. **Digestion** is the breakdown of food into molecules small enough for the body to absorb.
3. **Absorption** is the uptake of the small nutrient molecules by cells lining the digestive tract.
4. **Elimination** is the disposal of undigested materials left over from food.

Digestion: A Closer Look

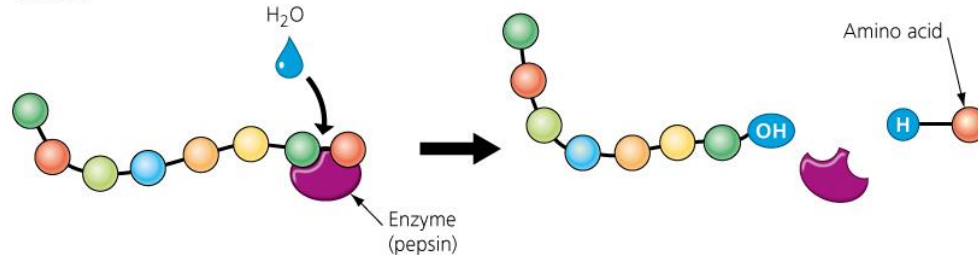
- The dismantling of food molecules is necessary because food molecules are too large to cross the membranes of animal cells and different from the molecules that make up an animal's body.
 - **Mechanical digestion** breaks chunks of food into small pieces, exposing them to **chemical digestion**.
 - Chemical digestion happens via **hydrolysis**, chemical reactions that break down large biological molecules by the addition of water molecules. Like most biological reactions, it requires **enzymes**.

From Cheese Protein to Human Protein

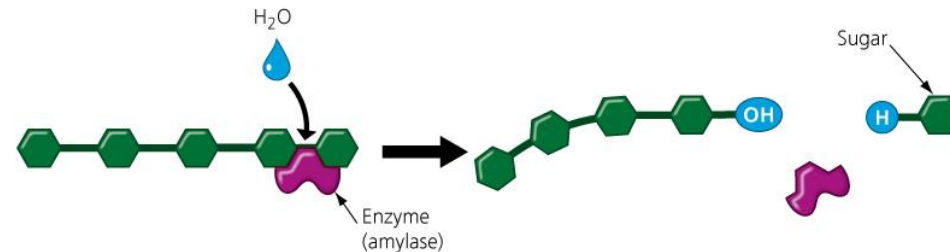


Chemical Digestion: Hydrolysis of Food Molecules

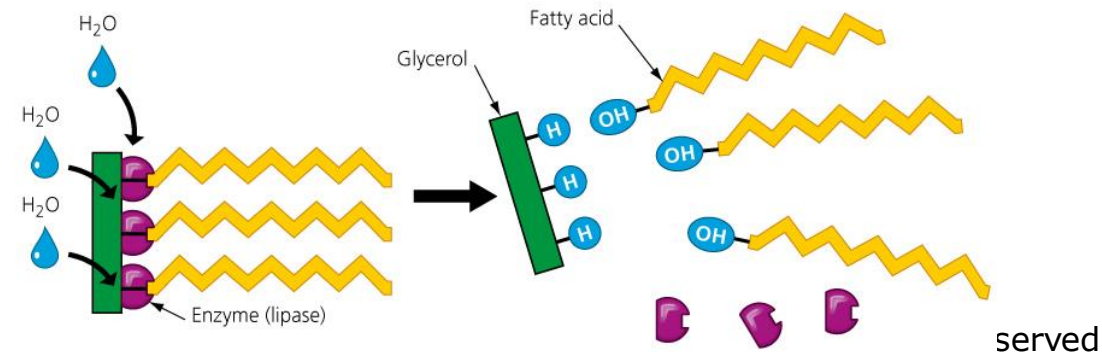
Protein



Carbohydrate



Fat



Digestive Compartments (1 of 2)

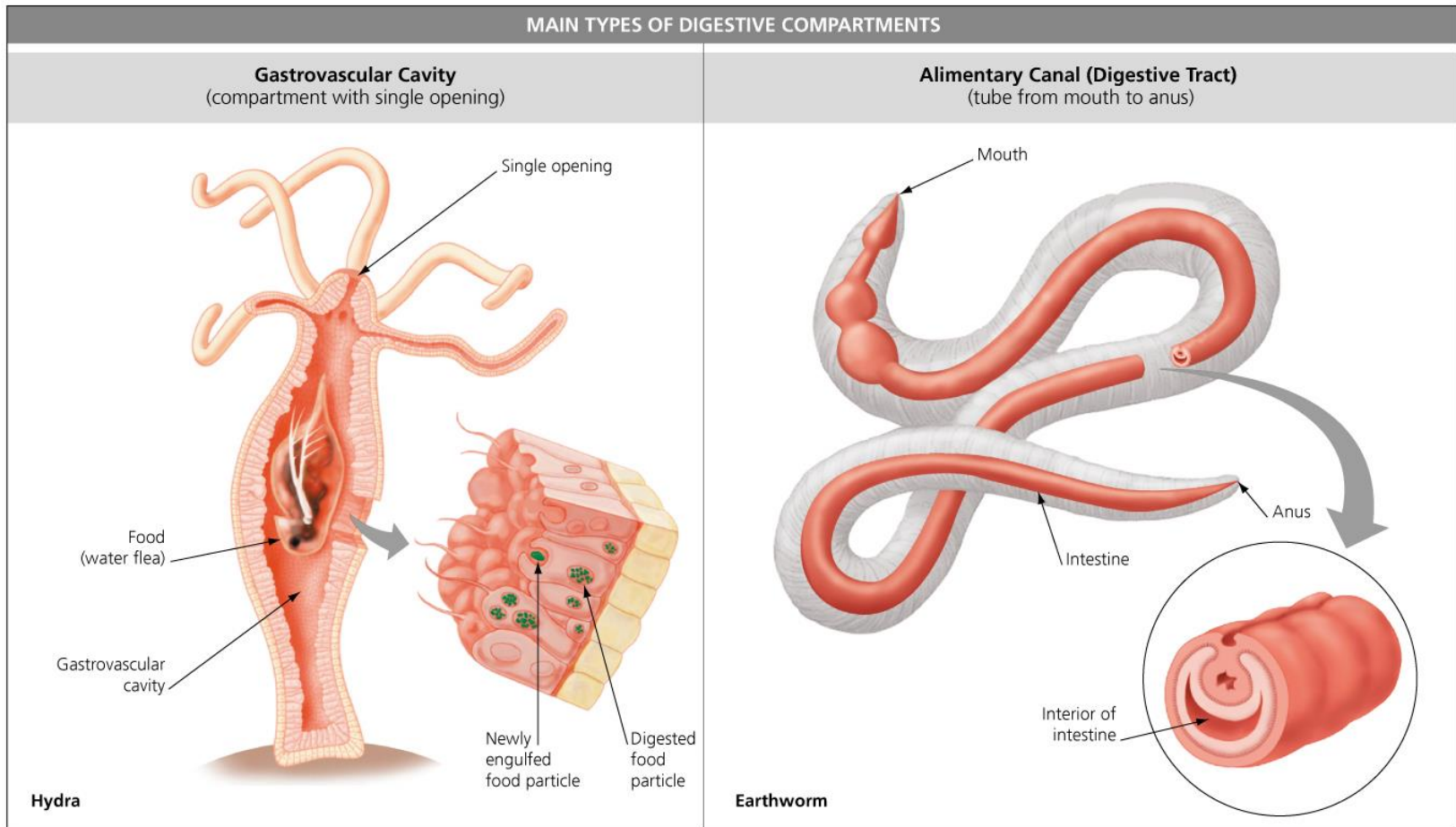
- How can an animal digest its food **without also digesting its own tissues**? Chemical digestion proceeds safely within some kind of compartment.
- The simplest type of digestion occurs within a cellular organelle.
- Most animals use a digestive compartment.
 - Simpler animals, including cnidarians and flatworms, have a **gastrovascular cavity**, a digestive compartment with a single opening that functions as the entrance for food (like a mouth) and the exit for undigested wastes (like an anus).

Digestive Compartments (2 of 2)

- The vast majority of animals, including earthworms and humans, have a digestive tube with two separate openings, a mouth at one end and an anus at the other.
 - Such a tube is called an **alimentary canal**, or **digestive tract**.
 - Food moves in just one direction through specialized regions of a digestive tube that digest and absorb nutrients in a stepwise fashion.

Alimentary: 영양의, 소화의

Main Types of Digestive Compartments in Animals



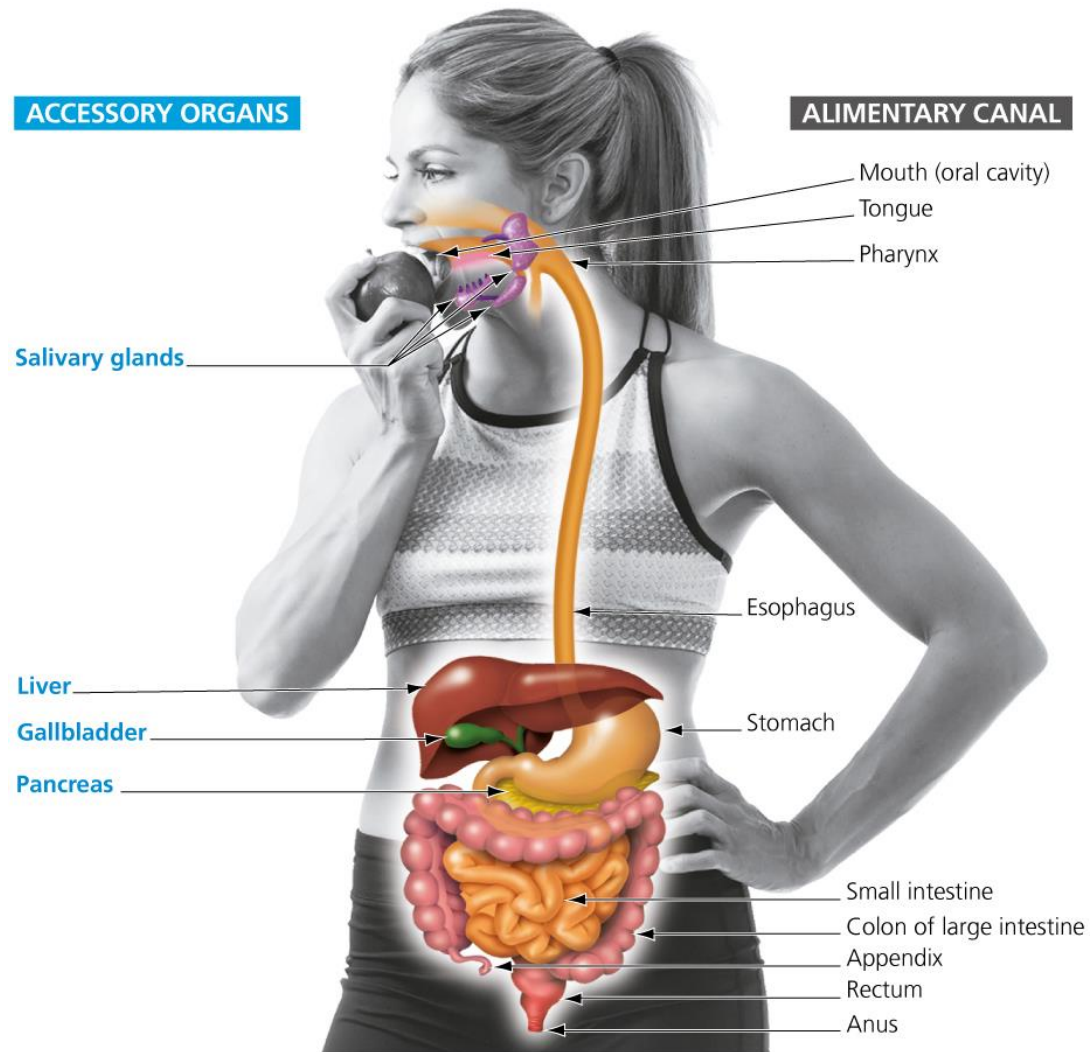
A Tour of the Human Digestive System: System Map

- We are now ready to follow a slice of pizza through the human alimentary canal, from mouth to anus.
- The human digestive system consists of an alimentary canal and several accessory organs (salivary glands, pancreas, liver, and gallbladder).
 - The accessory organs secrete digestive chemicals into the alimentary canal via ducts (thin tubes).
 - The alimentary canal is divided into specialized digestive organs along its length: mouth (oral cavity), pharynx, esophagus, stomach, small intestine, large intestine, and anus.

Pharynx: 인두

Esophagus: 식도

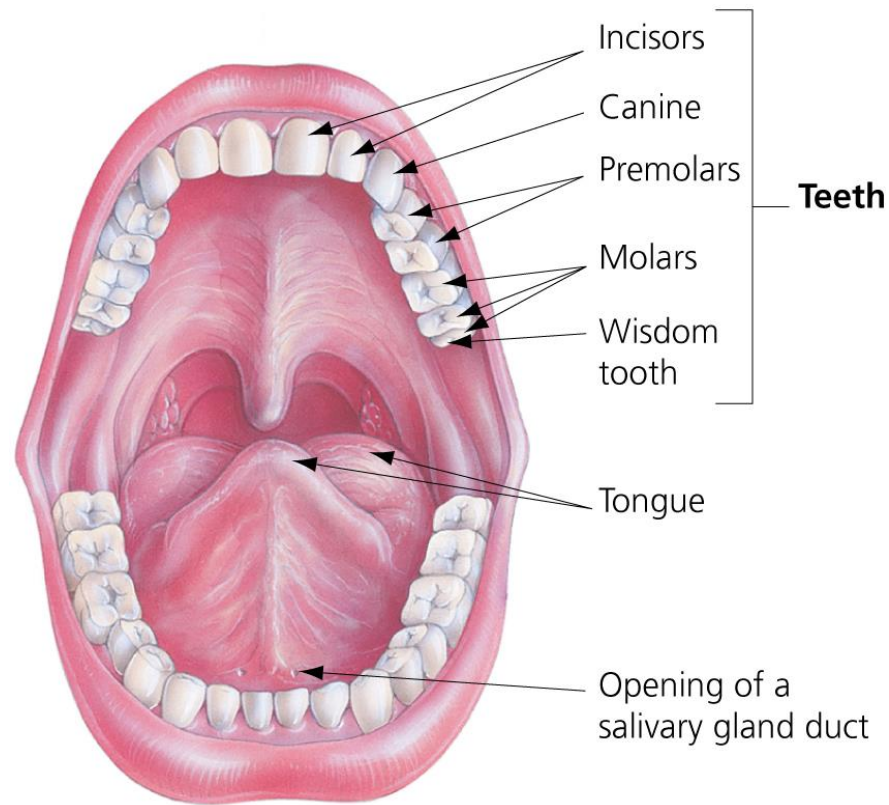
The Human Digestive System



The Mouth

- The **mouth**, also known as the **oral cavity**, is where food is ingested and where digestion starts.
 - Mechanical digestion begins here as the teeth cut, smash, and grind the food.
 - Chemical digestion begins in the mouth with the secretion of saliva from salivary glands.
 - The muscular **tongue** tastes, shapes food into a ball, and pushes the food to the back of the mouth for swallowing.

The Human Mouth and its Teeth

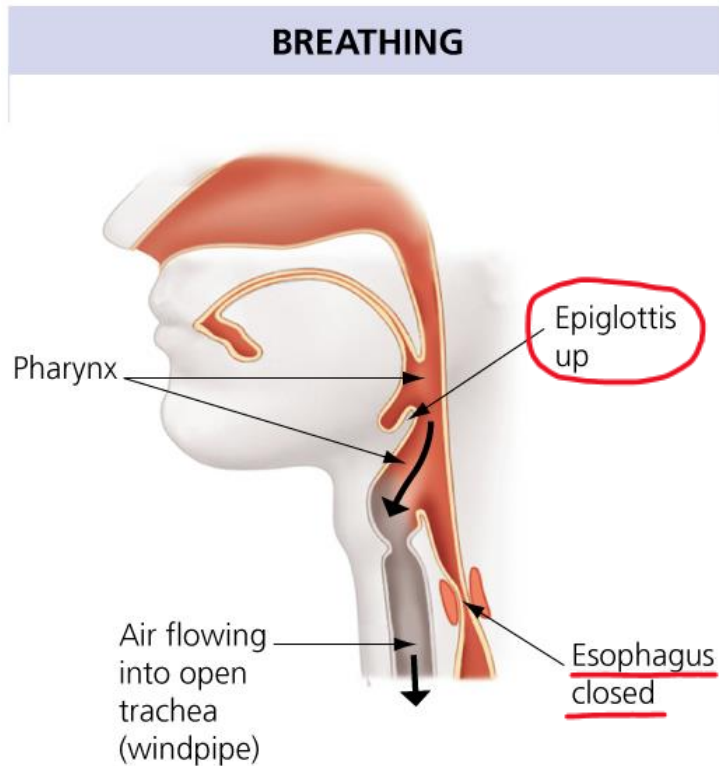


The Pharynx

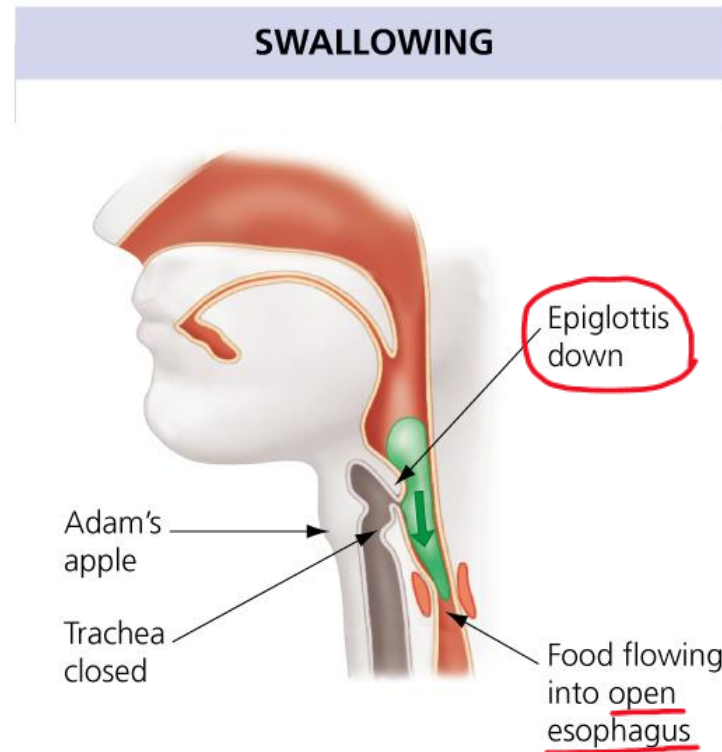
- The **pharynx**
 - connects the mouth to the esophagus (part of the digestive system) and
 - opens to the trachea, or windpipe, which leads to the lungs (part of the respiratory system).
- During swallowing, a reflex
 - moves the opening of the trachea upward and
 - tips a door-like flap called the epiglottis to close the trachea entrance.

Trachea: 기도
Windpipe: 기관
Epiglottis: 후두개

The Epiglottis Controls whether the Pharynx is Open to the Lungs (Left) or the Stomach (Right)



When you're not swallowing food or drink, air travels freely through the trachea (black arrows). The esophagus is closed.



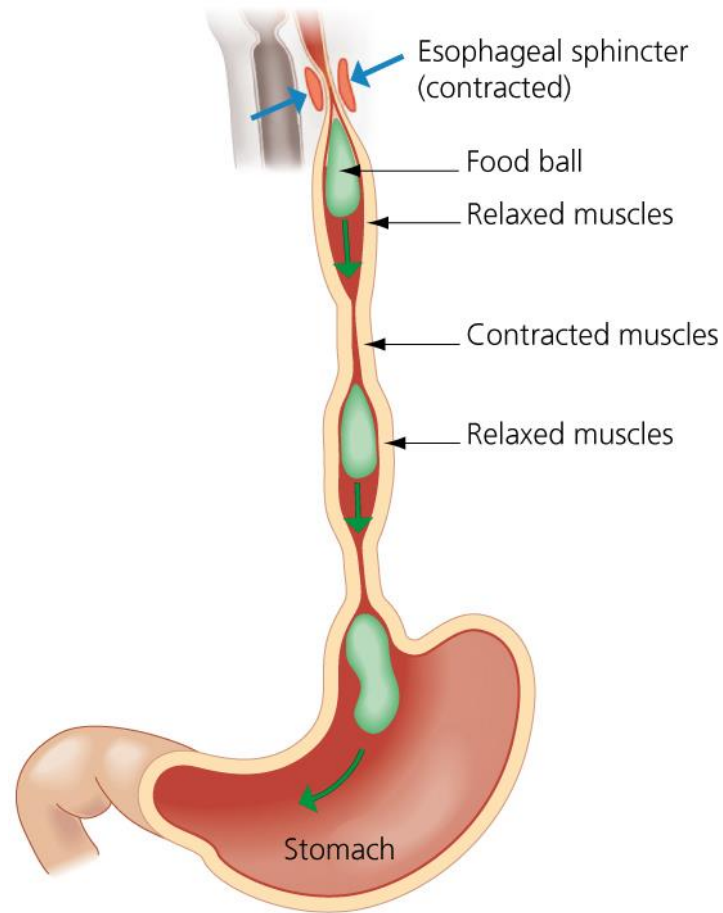
When the food reaches the back of the mouth, a swallowing reflex is triggered. The top of the trachea rises against the epiglottis, closing the air passage, while the esophagus opens. Food travels down the esophagus (green arrow).

The Esophagus

- The **esophagus**
 - is a muscular tube,
 - connects the pharynx to the stomach, and
 - moves food by **peristalsis**, alternating waves of muscular contraction and relaxation that squeeze the food ball along the esophagus.

Peristalsis: 연동운동

The Esophagus and Peristalsis



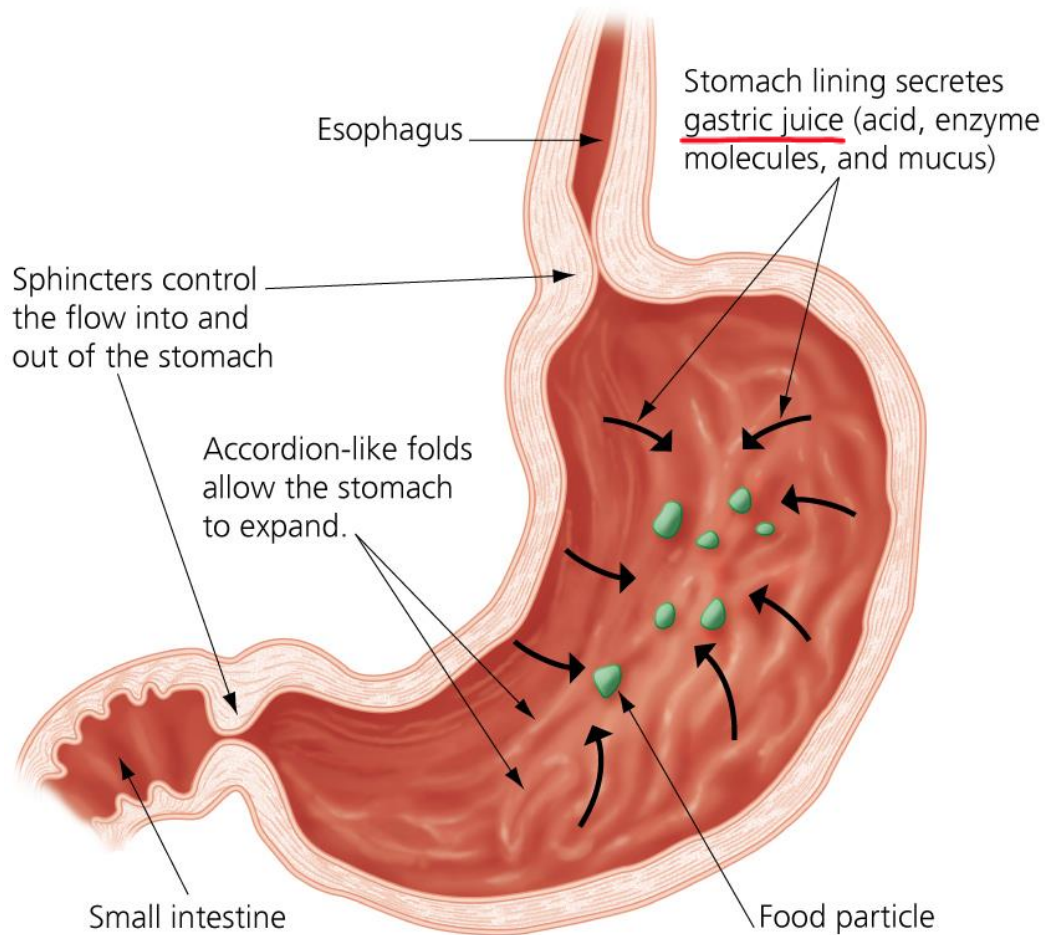
Sphincter: 괄약근

The Stomach (1 of 3)

- The human **stomach** is a large organ that acts as an expandable storage tank, holding enough food to sustain you for several hours.
- The stomach is a good example of the relationship between structure and function.
 - Its shape is like a collapsible canvas water bag that can be stretched out as needed, with an elastic wall and accordion-like folds.
 - This structure allows the stomach to store more than half a gallon of food and drink.



The Human Stomach



The Stomach (2 of 3)

- The cells lining the stomach's interior secrete a digestive fluid called **gastric juice**, made up of
 - **hydrochloric acid**,
 - digestive enzymes including **pepsin**, which breaks proteins into smaller pieces, and
 - **mucus**.
 - When food passes from the esophagus into the stomach, the muscular stomach walls begin to churn, mixing the food and gastric juice into a thick soup called **chyme**. The chyme leaves the stomach for the small intestine one squirt at a time.

Churn: 휘젓다

Chyme: 미즙

The Stomach (3 of 3)

- What keeps the stomach from digesting itself?
 - Mucus coating the stomach lining helps protect it from gastric juices and from abrasive materials in food.
 - Nerve and hormone signals regulate the secretion of gastric juice so that it is discharged only when food is in the stomach.
 - But gastric juice can still erode the stomach lining, requiring the production of new cells by cell division. Your stomach replaces its lining completely about once every three days!

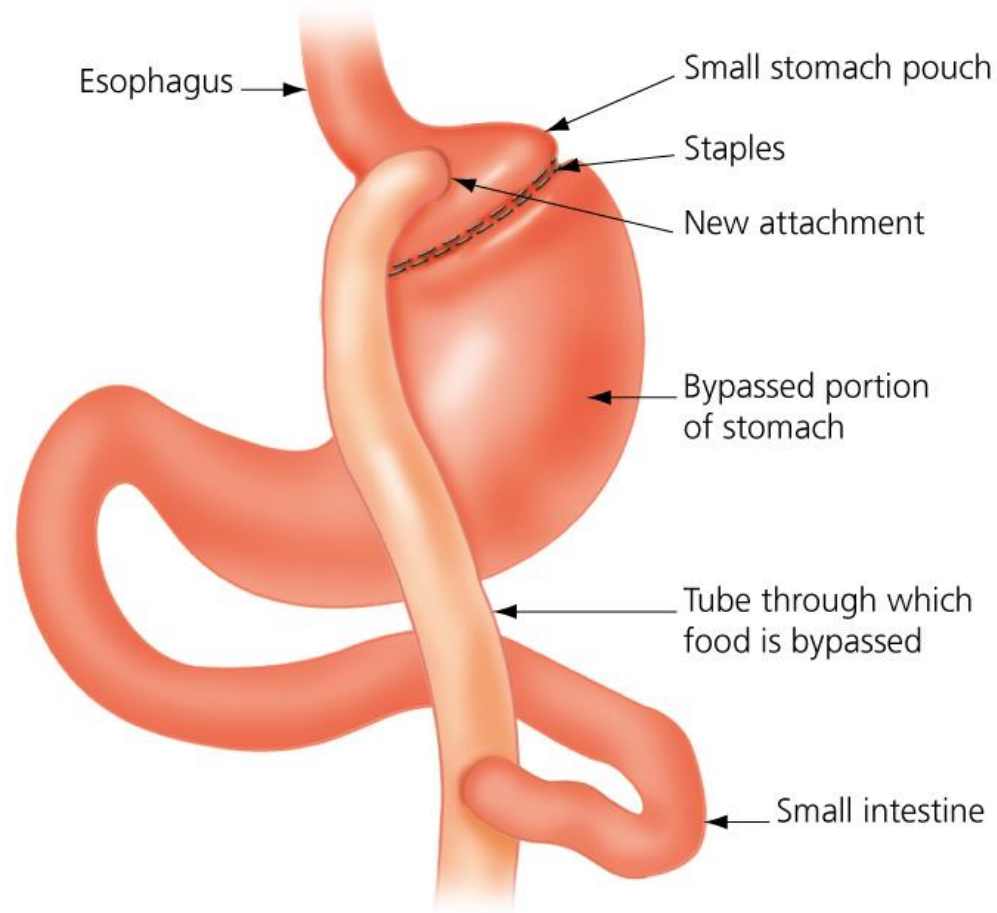
Stomach Ailments

- Heartburn is caused by backflow of chyme into the esophagus, commonly but inaccurately called “heartburn.”
 - Some people suffer this backflow frequently and severely enough to harm the lining of the esophagus, a condition known as **acid reflux or GERD (gastroesophageal reflux disease)**.
 - **Gastric ulcers** are erosions of the stomach lining and often caused by an acid-tolerant bacterium called *Helicobacter pylori*.

Weight Loss Surgeries

- The most common weight loss surgery in the United States is **gastric bypass**.
 - Staples are used to **reduce the stomach** to about the size of a chicken egg, and the **first 18 inches of the small intestine are bypassed** by attaching the downstream intestine directly to the reduced stomach pouch.
 - As a result, patients **quickly feel full** when eating and the body's **ability to absorb food is reduced**.
 - When accompanied by a healthy lifestyle, weight loss surgeries are successful in 90% of patients.

Gastric Bypass Surgery

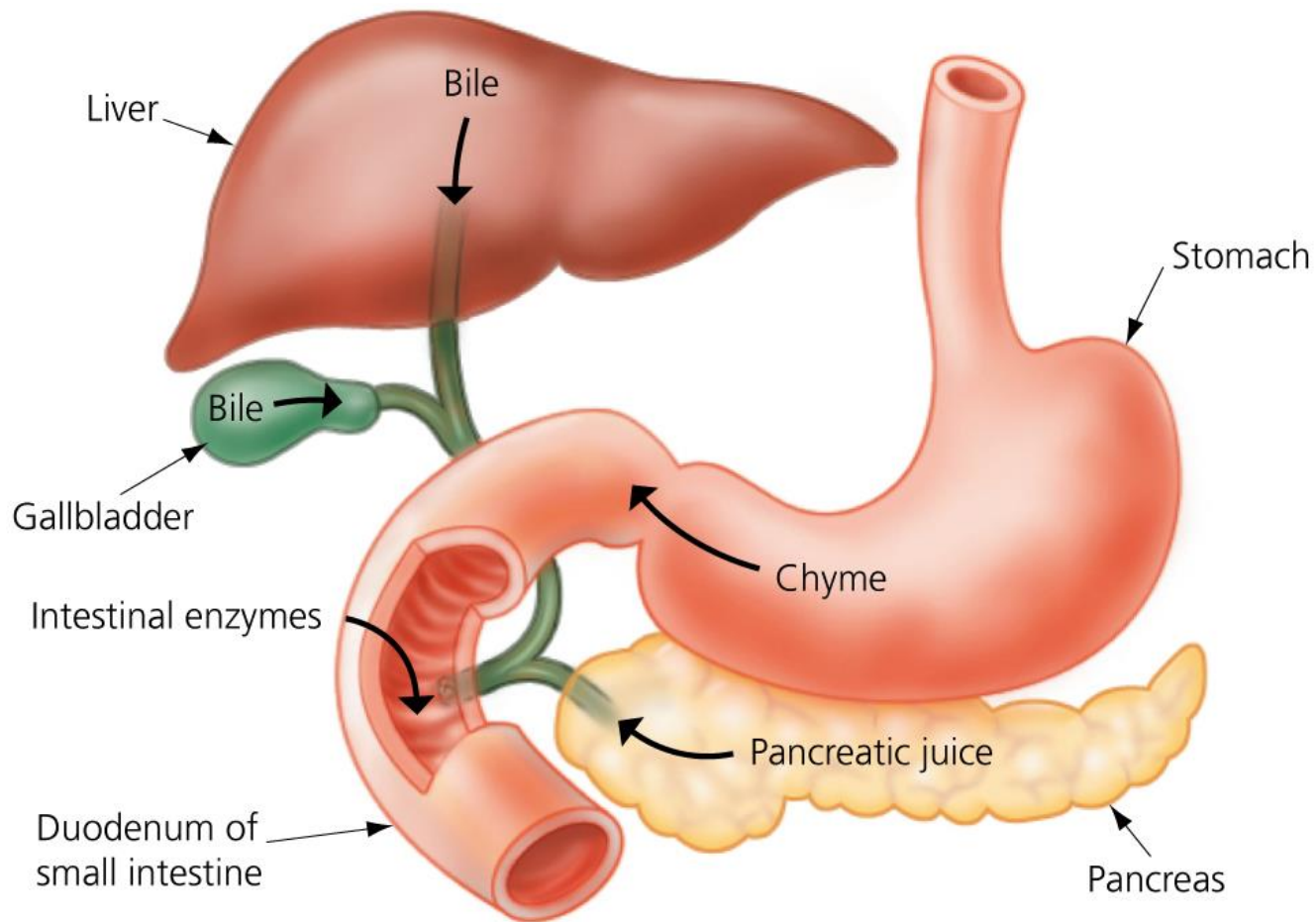


The Small Intestine: Chemical Digestion in the Small Intestine (1 of 2)

- The **small intestine** is
 - the longest part of the alimentary canal and
 - the major organ for chemical digestion and absorption of nutrients into the bloodstream.
- Enzymes are mixed with chyme in the first 25 cm or so (about a foot) of the small intestine, the region called the **duodenum**.
- The duodenum receives digestive juices from the pancreas, liver, gallbladder, and intestinal lining.

Duodenum: 십이지장
Duodenum digitorium

The Duodenum



Chemical Digestion in the Small Intestine

(1 of 3)

- The **pancreas** is a large gland that secretes pancreatic juice into the duodenum via a duct.
- Pancreatic juice
 - neutralizes the stomach acid that enters the duodenum and
 - contains enzymes that aid in digestion.
- As peristalsis propels the mix along the small intestine, these enzymes contribute to the breakdown of food molecules.

Chemical Digestion in the Small Intestine

(2 of 3)

- **Bile** is a juice
 - produced by the **liver**,
 - stored in the **gallbladder**, and
 - secreted through a duct into the duodenum.
- Bile contains salts that break up fats into small droplets that are more susceptible to dismantling by digestive enzymes.

Bile: 담즙

Chemical Digestion in the Small Intestine

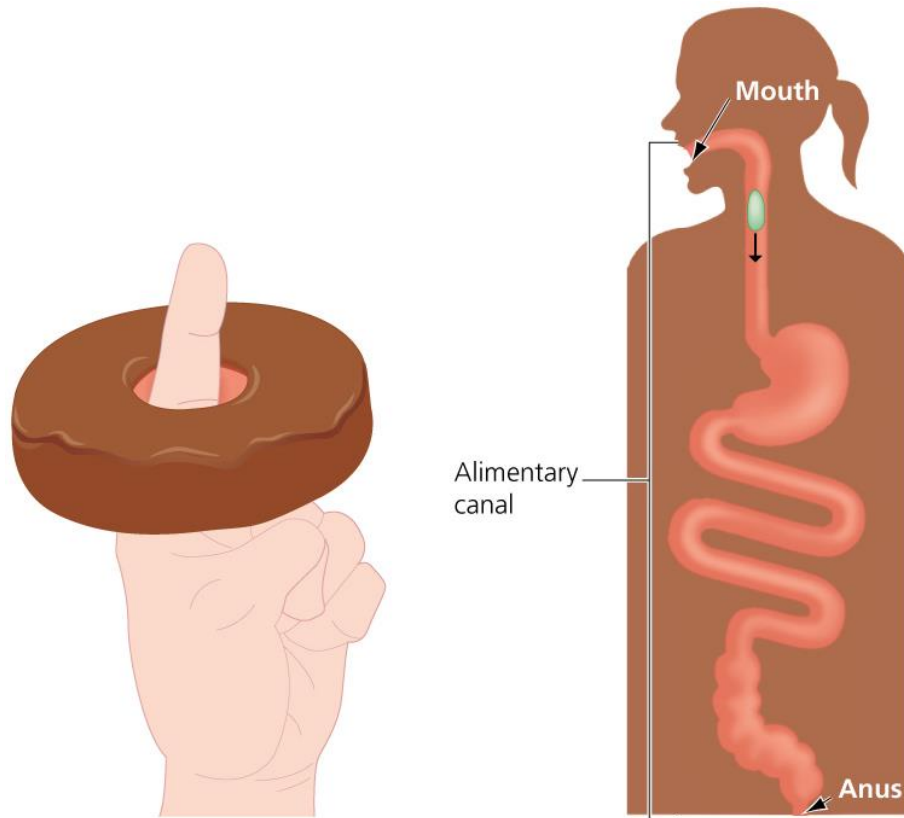
(3 of 3)

- In addition to digesting fats, the liver performs several other vital functions.
 - Capillaries from the small and large intestines converge into veins (portal vein) that lead into the liver. The liver can detoxify the blood. For example,
 - it converts alcohol into inactive substances that are secreted in the urine and
 - converts nutrients into new substances, several of which regulate other metabolic processes.
 - The liver's location between the intestines and the heart provides a good illustration of how interactions between body systems keep us healthy.

Absorption of Nutrients (1 of 2)

- The alimentary canal is a tube running through the body, and its cavity is continuous with the great outdoors.
 - Until nutrients actually cross the tissue lining the alimentary canal and enter the bloodstream, they are still outside the body.
 - If it were not for nutrient absorption, we could eat and digest huge meals but still starve.

Nutrients within the Small Intestine are **not** yet Inside the Body



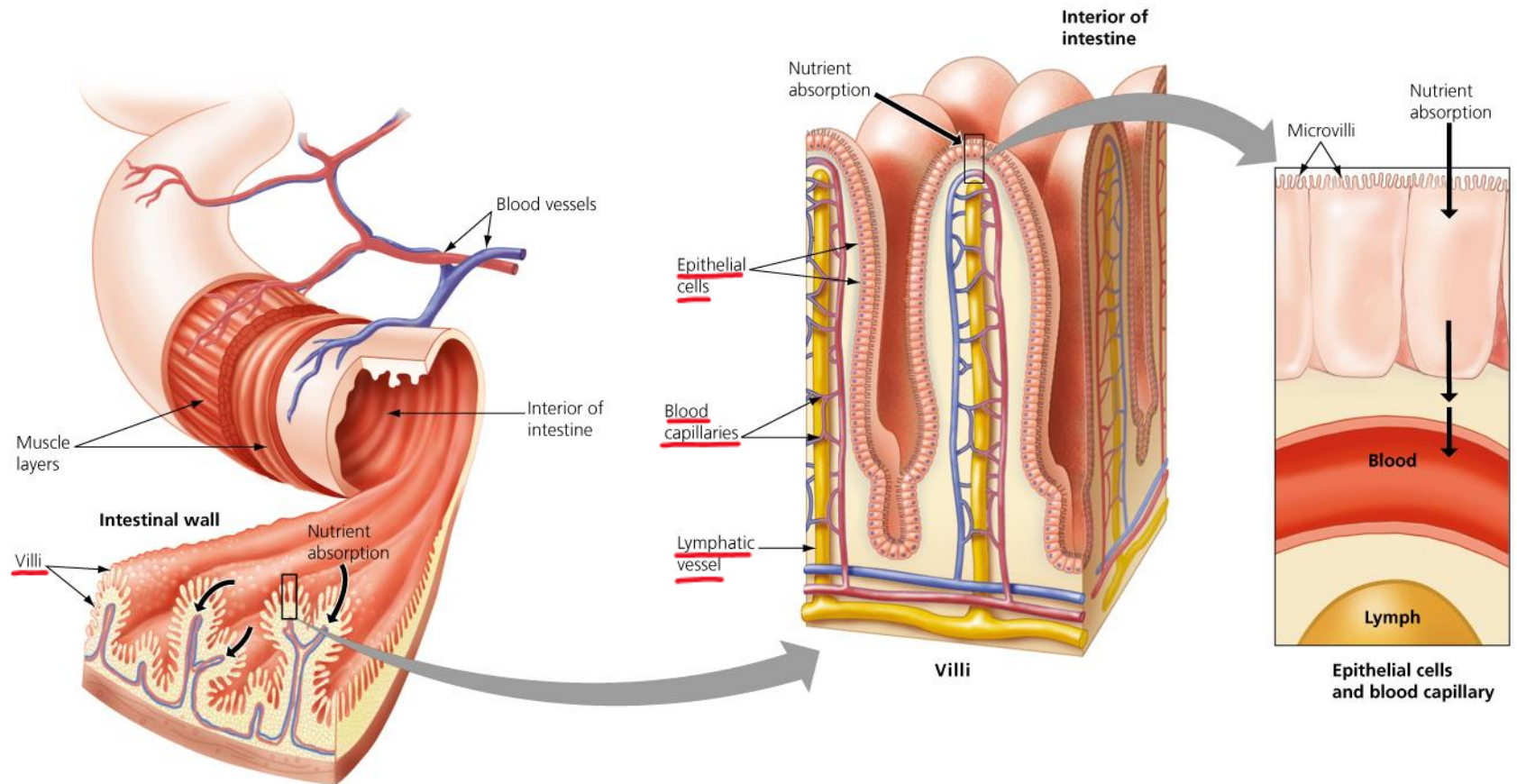
A finger through a hole

Food through the alimentary canal

Absorption of Nutrients (2 of 2)

- The structure of the intestinal lining fits its function.
 - The structure of the **epithelium**, with its expansive surface area, is an evolutionary adaptation that correlates with the function of this part of the alimentary canal: the absorption of nutrients.
 - Absorbed small molecules pass from the digestive tract into the network of small blood vessels and lymphatic vessels in the core of each villus.
 - After nutrients have crossed the cell membranes of the microvilli, they are finally inside the body, where the bloodstream and lymph carry them away to distant cells.

The Small Intestine and Nutrient Absorption



The Human Microbiome

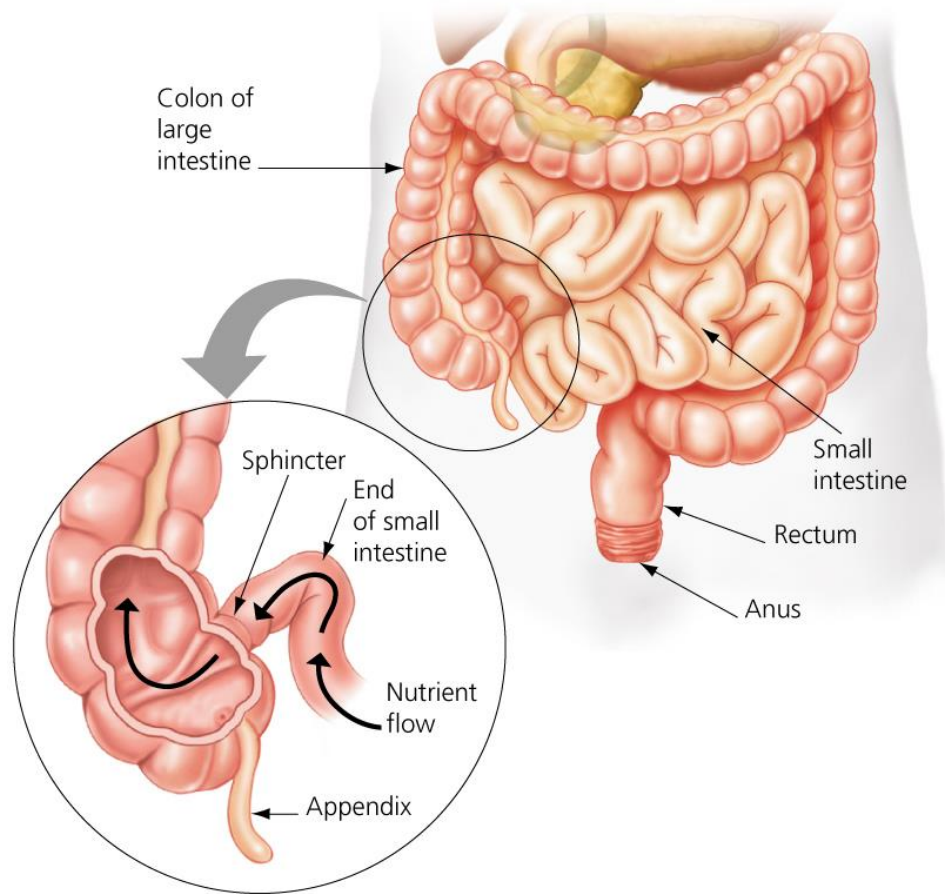
- In recent years, medical researchers have greatly expanded our knowledge of the more than 400 species of bacteria that constitute the **microbiome** (bacterial community) that lives in the human digestive system.
- Unlike the harmful bacteria that cause ulcers or cholera, most of our microbiome illustrates the principle of mutualistic symbiosis, an interaction between two species that benefits both. For example, some intestinal bacteria produce vitamins B and K and regulate the development of the intestinal lining.

The Large Intestine (1 of 4)

- The **large intestine** is shorter than the small intestine but almost twice as wide and about 1.5 meters in length.
 - At the junction of the small and large intestine is a small, finger-like extension called the **appendix**.
 - **Appendicitis** is a bacterial infection of the appendix.
- The **colon**
 - forms the main portion of the large intestine,
 - absorbs water from the alimentary canal, and
 - produces feces, consisting of undigested material.

Colon: 결장

The Large Intestine and its Connection to the Small Intestine



The Large Intestine (2 of 4)

- If the lining of the colon is irritated by a viral or bacterial infection, the colon may be unable to reabsorb water efficiently, resulting in diarrhea.
 - Prolonged diarrhea can cause life-threatening dehydration, particularly among the very young and very old.
 - Constipation occurs when peristalsis moves feces along too slowly and the colon reabsorbs so much water that the feces become too compacted. It can result from lack of exercise or a diet that does not include enough plant fiber.

The Large Intestine (3 of 4)

- Several intestinal disorders are characterized by inflammation (painful swelling) of the intestinal wall.
 - Celiac disease results when gluten, a protein found in wheat, triggers an immune reaction that leads to swelling and a lack of nutrient absorption.
 - An inappropriate immune response causes Crohn's disease, a chronic inflammation that can periodically flare up along any part of the alimentary canal.

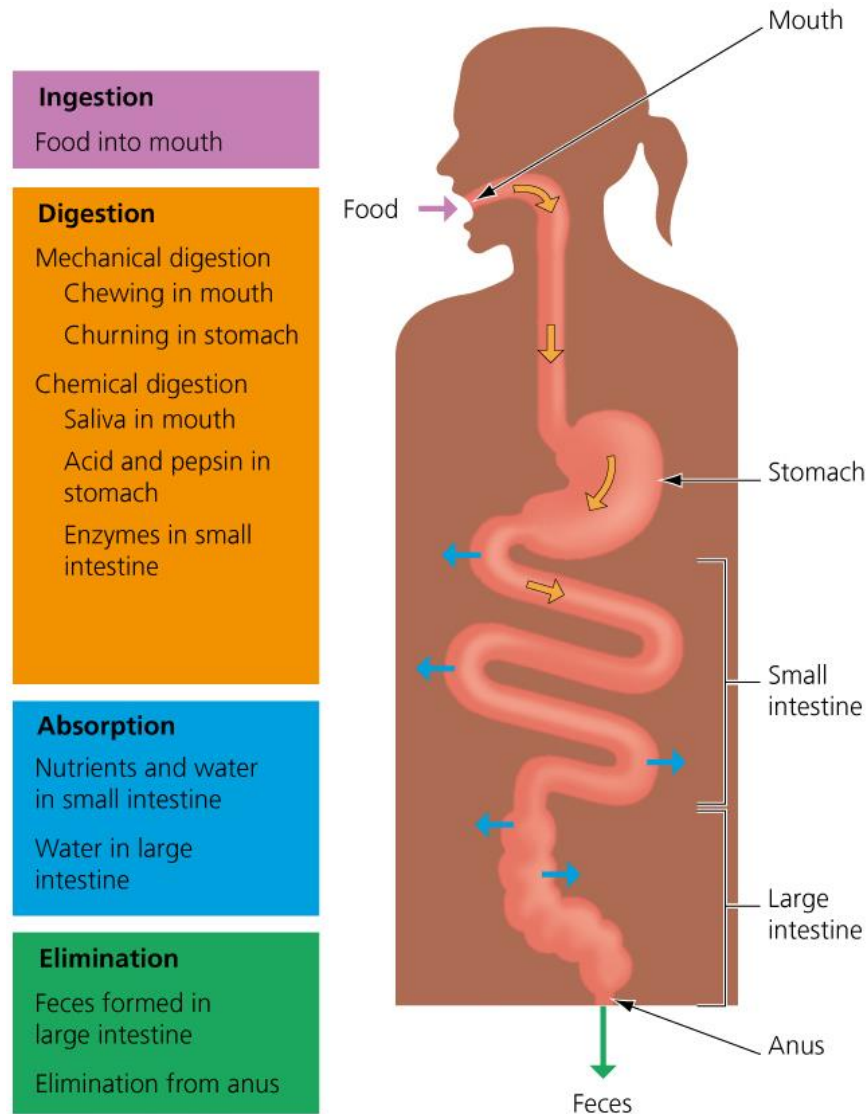
Celiac disease: 만성소화장애

The Large Intestine (4 of 4)

- The **rectum**, the last 15 cm (6 inches) of the large intestine, stores feces until they can be eliminated.
 - Two rectal sphincters, one voluntary and the other involuntary, regulate the opening of the anus.
 - When the voluntary sphincter is relaxed, contractions of the rectum expel feces.
- **Figure 22.15** stretches out the alimentary canal to help you review food processing along its length.

Rectum: 직장

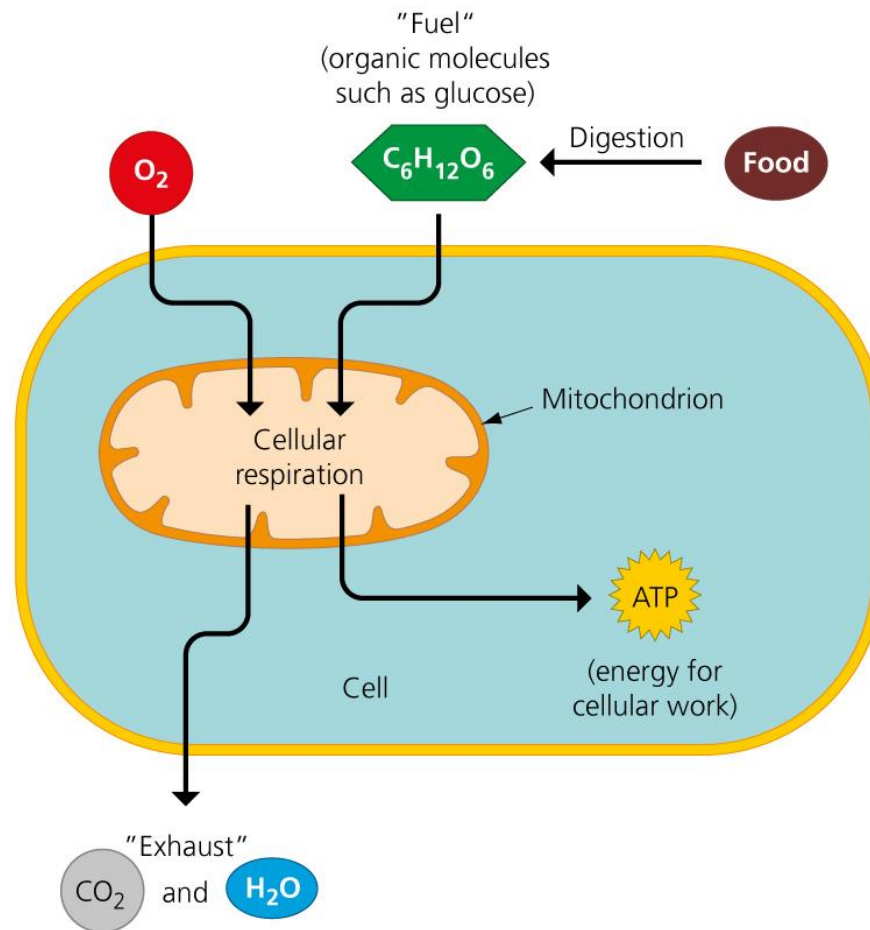
Review of Food Processing in the Human Alimentary Canal



Human Nutritional Requirements: Food as Fuel

- Proper nutrition provides fuel for cellular work, materials for building molecules, and essential nutrients for health.
- One of biology's overarching themes is the transformation of energy and matter.
 - All living cells take in energy, convert it to useful forms, and expel energy.
 - Cells can extract energy stored in the organic molecules of food through the process of cellular respiration and expend that energy as cellular work.

Review of Cellular Respiration



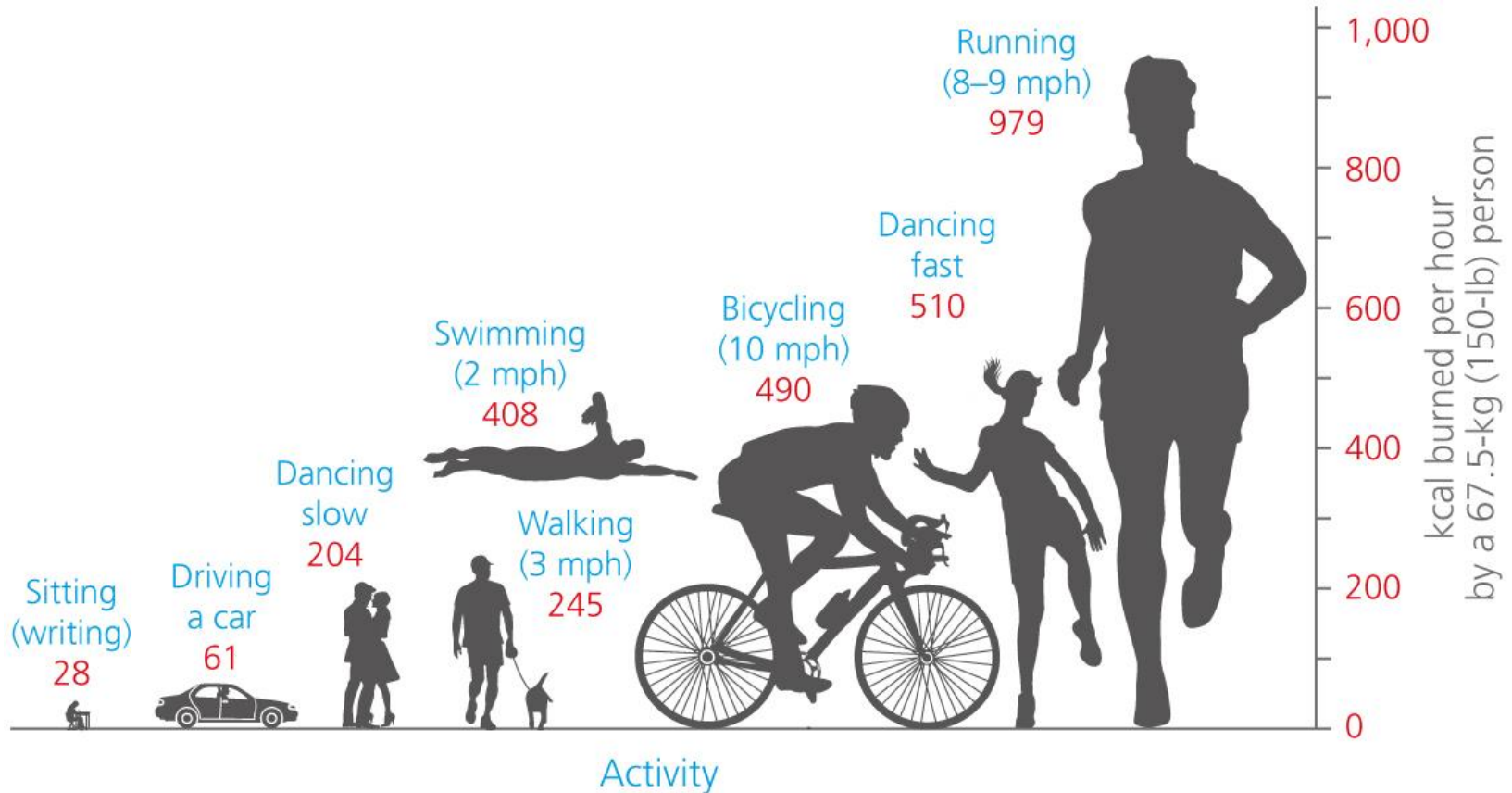
Calories

- **Calories** are a measure of the energy stored in your food and used during daily activities.
 - One calorie is the amount of energy required to raise the temperature of one gram of water by 1°C.
 - A **kilocalorie (kcal)** is 1,000 calories, the unit listed on food labels, and often called a Calorie, with an uppercase C.

Metabolic Rate

- The rate at which your body consumes energy is called your **metabolic rate**.
 - Your overall metabolic rate is equal to your **basal metabolic rate (BMR)**, the amount of energy it takes just to maintain your basic body functions, plus any energy consumption above that base rate.
 - The more active you are, the greater your actual metabolic rate and the greater the number of calories your body uses per day.
 - Metabolic rate also depends on other factors, such as body size, age, stress level, and heredity.

Energy (kcal) Burned per Hour by a 150-Pound Person Performing Various Activities



Food as Building Material

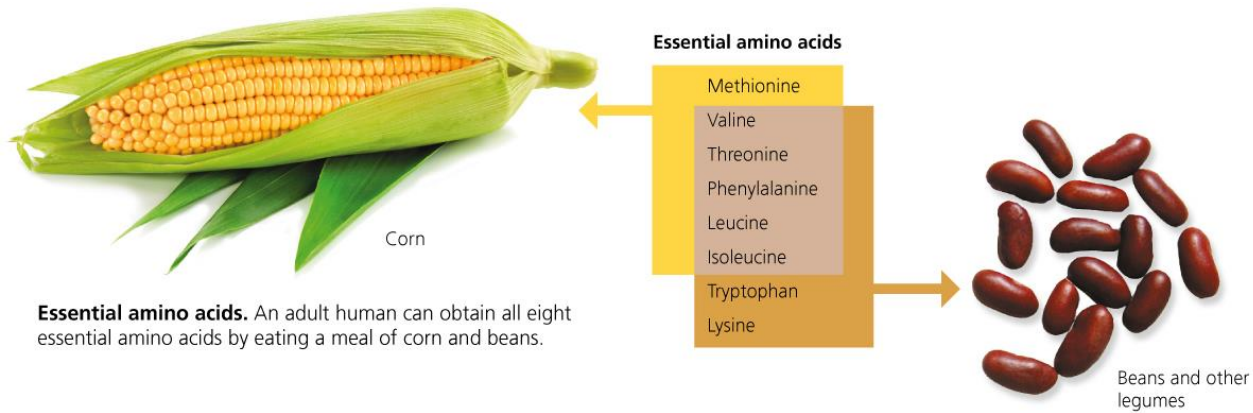
- Your health depends on the continuous repair and maintenance of your tissues.
 - This requires small organic molecules (monomers) produced during the digestion of food. Your cells can reassemble those smaller molecules into various large biological molecules.
 - Within limits, your metabolism can change organic material from one form to another to compensate for nutrients that are lacking in your diet.
 - However, **essential nutrients** cannot be made from any other materials, so the body needs to receive them in preassembled form.

Essential Amino Acids

- All proteins are built from 20 different kinds of amino acids.
 - In humans, **eight essential amino acids** must be obtained from the diet and are in different proportions in different foods.
 - All eight essential amino acids can be consumed by eating meat, eggs, or milk or the combination of grains and legumes such as beans, peanuts, and peas.

Legume: 콩과 식물

Essential Amino Acids from a Vegetarian Diet



Rice and beans



Bread and peanut butter



Pita bread and hummus

Complete meals. Many societies have a staple meal that provides a complete set of essential amino acids by combining a grain with a legume.

Vitamins (1 of 2)

- **Vitamins** are organic molecules required in the diet in very small amounts. **Most vitamins assist enzymes.**
- There are 13 vitamins essential to human health.
 - People who eat a balanced diet should be able to obtain enough of all needed nutrients in their food.
 - For others, vitamin supplements can fill in the gaps, but supplements should not be used indiscriminately because overdoses of some vitamins (such as A, D, and K) can be harmful.

Vitamins (2 of 2)

Table 22.1 Vitamins		
Vitamin	Major Dietary Sources	Interesting Facts
Water-Soluble Vitamins		
Vitamin B ₁ (thiamine)	Pork, legumes, peanuts, whole grains	Too little thiamin leads to the disease beriberi; refined grains (like polished white rice) lack thiamin.
Vitamin B ₂ (riboflavin)	Dairy products, organ meats, enriched grains, vegetables	Deficiency causes photophobia (aversion to light).
Niacin	Nuts, meats, fish, grains	High doses reduce cholesterol but can cause hot flashes. Too little leads to the disease pellagra, which can be deadly.
Vitamin B ₆ (pyridoxine)	Common in most foods: meats, vegetables, whole grains, milk, legumes	Deficiency is rare.
Pantothenic acid	Most foods: meats, fish, dairy, whole grains	Component of coenzyme A. Deficiency is rare but can lead to fatigue.
Folic acid (folate)	Green leafy vegetables, oranges, nuts, legumes, whole grains, fortified foods	Recommended as a supplement for women of childbearing age because it cuts in half the risk of some birth defects involving development of the spinal cord.
Vitamin B ₁₂	Animal products: meats, eggs, dairy	It's hard for vegans to get enough B ₁₂ in their diet; some intestinal disorders (such as Crohn's disease or celiac disease) may cause deficiencies.
Biotin	Legumes, most vegetables, meats, milk, liver, egg yolks	For most people, adequate levels are provided by intestinal bacteria.
Vitamin C (ascorbic acid)	Raw fruits and vegetables, especially citrus fruits, broccoli, cabbage, tomatoes, green peppers, strawberries	Many animals can produce their own vitamin C, but humans cannot. Deficiency causes the disease scurvy, which was a significant health problem during the era of lengthy sea voyages.
Fat-Soluble Vitamins		
Vitamin A	Dark green and orange vegetables and fruits, dairy products	Too little in the diet can lead to vision loss. Too much can cause yellow/orange skin and liver damage.
Vitamin D	Fortified dairy products, egg yolk	Made in human skin in presence of sunlight; aids calcium absorption and bone formation. Too little causes rickets (bone deformities).
Vitamin E (tocopherol)	Green leafy vegetables, oils, nuts, seeds, wheat germ	Antioxidant that helps prevent damage to cell membranes. Deficiency is rare.
Vitamin K	Green vegetables, vegetable oils	Needed for blood clotting. Produced by intestinal bacteria, so newborns and people taking long-term antibiotics may be deficient.

Minerals

- We also require smaller amounts of 21 other chemical elements that are acquired mainly in the form of inorganic nutrients called **minerals**.
- Like vitamin deficiencies, mineral deficiencies or overdoses can cause health problems.
 - Too little calcium can result in the degenerative bone disease osteoporosis.
 - Excess sodium can contribute to high blood pressure.

Essential Fatty Acids

- Our cells make fats and other lipids by combining fatty acids with other molecules, such as glycerol.
 - **Essential fatty acids** must be obtained in the diet.
 - Most diets furnish ample amounts of the essential fatty acids, so deficiencies are rare.

Omega-3, -6 fatty acids

Decoding Food Labels

- The U.S. Food and Drug Administration (FDA) requires two blocks of information on labels:
 - the ingredients (by weight) from greatest amount to least and
 - key nutrients, emphasizing the ones associated with disease and the ones associated with a healthy diet.
- Reading food labels carefully can help you make informed choices about what you put into your body.

A Food Label in the Previous Format (Left) and the Revised Format (Right)

Nutrition Facts			
Serving Size 2/3 cup (55g)			
Servings Per Container About 8			
Amount Per Serving			
Calories 230		Calories from Fat 72	
		% Daily Value*	
Total Fat	8g		10%
Saturated Fat	1g		5%
Trans Fat	0g		
Cholesterol	0mg		0%
Sodium	160mg		7%
Total Carbohydrate	37g		13%
Dietary Fiber	4g		14%
Sugars	12g		
Protein	3g		
Vitamin A			10%
Vitamin C			8%
Calcium			20%
Iron			45%
*Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:			
	Calories:	2,000	2,500
Total Fat	Less than	65g	80g
Sat Fat	Less than	20g	25g
Cholesterol	Less than	300mg	300mg
Sodium	Less than	2,400mg	2,400mg
Total Carbohydrate		300g	375g
Dietary Fiber		25g	30g

Nutrition Facts	
8 servings per container	
Serving size 2/3 cup (55g)	
Amount per serving	
Calories	230
% Daily Value*	
Total Fat 8g	10%
Saturated Fat 1g	5%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 160mg	7%
Total Carbohydrate 37g	13%
Dietary Fiber 4g	14%
Total Sugars 12g	
Includes 10g Added Sugars	20%
Protein 3g	
Vitamin A 8mg	10%
Vitamin C 5mg	8%
Calcium 260mg	20%
Iron 8mg	45%
*The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

Nutritional Disorders: Malnutrition

- Dietary problems can have severe health consequences. **Malnutrition** refers to problems caused by an improper or insufficient diet.
- Protein deficiency is most common in less industrialized countries where there is a great gap between food supply and population size.
 - People who get almost all their calories from a single plant staple, such as corn or potatoes, will suffer deficiencies of essential amino acids.
 - Most victims of protein deficiency are children, who may develop poorly both physically and mentally.

Eating Disorders

- Millions of Americans, mostly female, are affected by malnutrition not because they lack access to food, but because they have an eating disorder.
 - **Anorexia nervosa** is characterized by self-starvation due to an intense fear of gaining weight, even when the person is underweight.
 - **Bulimia** is a behavioral pattern of binge eating followed by purging through induced vomiting, abuse of laxatives, or excessive exercise.
 - Both disorders are characterized by an obsession with body weight and shape and can result in serious health problems.

Anorexia nervosa: 거식증

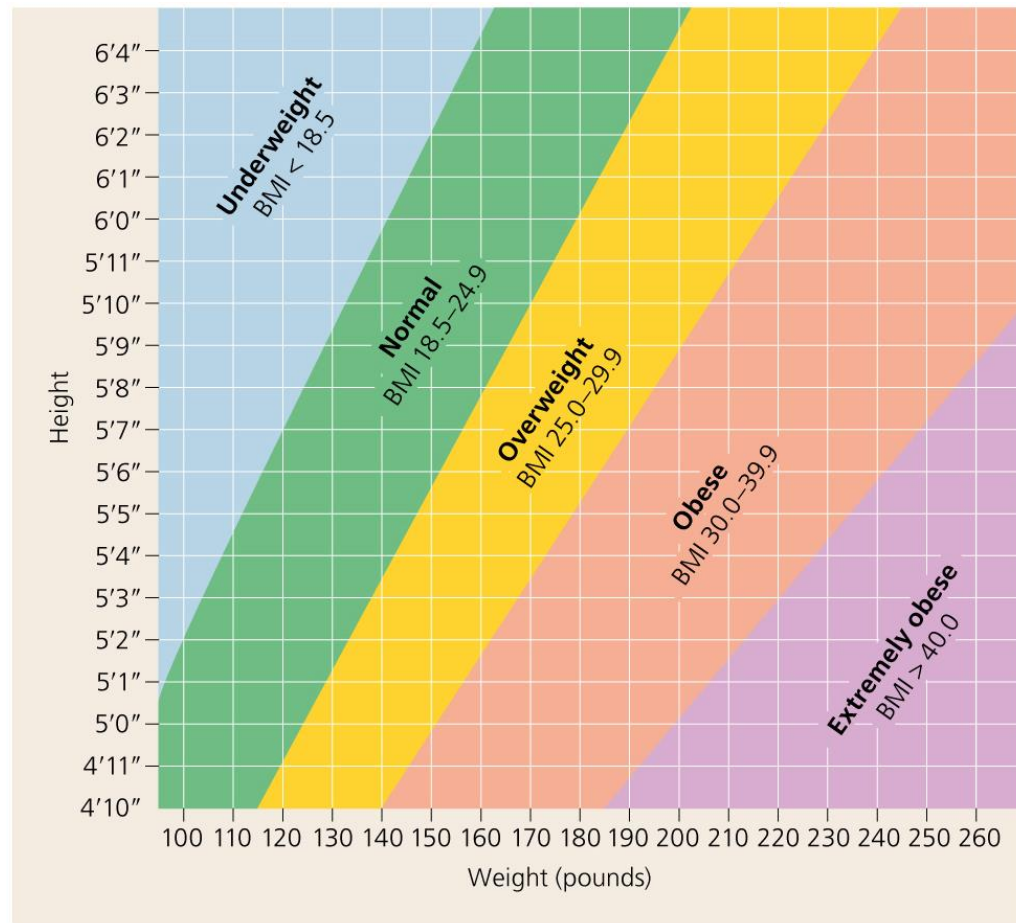
Bulimia: 식욕 이상 항진증 (폭식 후 토하기 반복)

Laxative: 설사약

Obesity

- **Obesity**
 - is a too-high **body mass index (BMI)**,
 - is the nutritional disorder of greatest concern,
 - affects about one-third of all Americans, and
 - increases the risk of heart attack, diabetes, cancer, and several other diseases.
- BMI charts show a range of acceptable values.

Body Mass Index (BMI): One Measure of Healthy Weight



The Process of Science: Can a Gene Make You Fat? (1 of 2)

- **Background:** A mutation in a particular gene in mice leads to a significant increase in body fat. But how does a small change in DNA cause such a large change in the body? Researchers formed the hypothesis that mutant mice become overweight because their obese gene fails to produce a protein called **leptin**. If this is true, then injecting leptin into mutant mice would overcome the effects of the defective obese gene.
- **Method:** Researchers injected some mutant and normal mice with leptin (the experimental groups) and injected others with saline (the control groups).

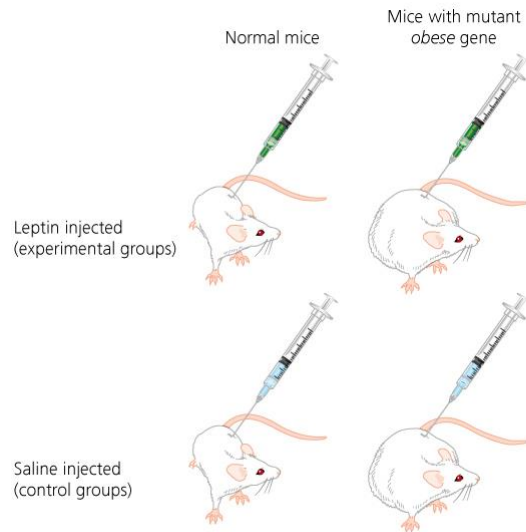
The Process of Science: Can a Gene Make You Fat? (2 of 2)

- **Results:** Mutant mice that received leptin ended the study, weighing about half as much as those that received saline. Normal mice receiving leptin lost weight, compared with those receiving saline, but the difference was not as large.
- More than 100 genes are known to contribute to weight maintenance in humans.
- Genetics only partially explains why certain people must fight hard to control their weight.

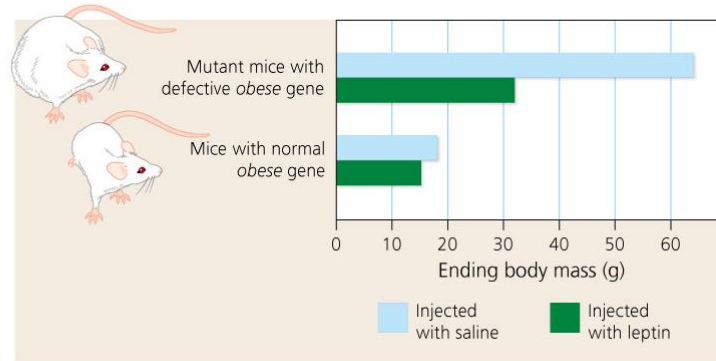
Investigating the genetic basis of obesity



(a) An obese mouse (left) with a defective gene that normally produces an appetite-regulating protein



(b) Testing the effect of the hormone leptin



(c) The effect of leptin on mice with the *obese* gene

Evolution Connection: Fat and Sugar Cravings

- Most Americans crave fatty or sweet foods. Why do we hunger for unhealthy foods?
- For most of human history, our ancestors were continually in danger of starvation.
 - In such a feast-or-famine existence, natural selection may have avored individuals who gorged themselves on the rare occasions when rich, fatty foods were available.
 - With their ample reserves, they were more likely than thinner peers to survive the inevitable famines.

Gorge: 잔뜩 먹다

Modern Hunter-Gatherers

Hunter-gatherer: 수렵 채집인



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